EXPERIMENT – 7

AIM :
To identify functional dependencies in a denormalized Sales table and normalize it step-by-step from UNF to 3NF and BCNF. To verify lossless join and dependency preservation using SQL queries.

SOURCE CODE :

```
-- Clean up tables so this script can be run again
DROP TABLE IF EXISTS OrderDetails;
DROP TABLE IF EXISTS Orders;
DROP TABLE IF EXISTS Products;
DROP TABLE IF EXISTS Customers;
DROP TABLE IF EXISTS Sales;

-- Create the original denormalized sales table
CREATE TABLE Sales (
    OrderID INT,
    CustomerID INT,
    CustomerName VARCHAR(50),
    ProductID INT,
    ProductName VARCHAR(50),
    Quantity INT,
    Price DECIMAL(10,2)
);

INSERT INTO Sales (OrderID, CustomerID, CustomerName, ProductID, ProductName,
Quantity, Price) VALUES
(1, 101, 'Amit', 201, 'Laptop', 2, 50000.00),
(1, 101, 'Amit', 202, 'Mouse', 3, 500.00),
(2, 102, 'Ravi', 201, 'Laptop', 1, 50000.00);

-- Create normalized tables
CREATE TABLE Customers (
    CustomerID INT PRIMARY KEY,
    CustomerName VARCHAR(50)
);

CREATE TABLE Orders (
    OrderID INT PRIMARY KEY,
    CustomerID INT NOT NULL,
    FOREIGN KEY (CustomerID) REFERENCES Customers(CustomerID)
);

CREATE TABLE Products (
    ProductID INT PRIMARY KEY,
```

```
    ProductName VARCHAR(50),
    Price DECIMAL(10,2)
);

CREATE TABLE OrderDetails (
    OrderID INT,
    ProductID INT,
    Quantity INT,
    PRIMARY KEY (OrderID, ProductID),
    FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
    FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);

-- Load the normalized tables from Sales
INSERT INTO Customers (CustomerID, CustomerName)
SELECT DISTINCT CustomerID, CustomerName
FROM Sales;

INSERT INTO Orders (OrderID, CustomerID)
SELECT DISTINCT OrderID, CustomerID
FROM Sales;

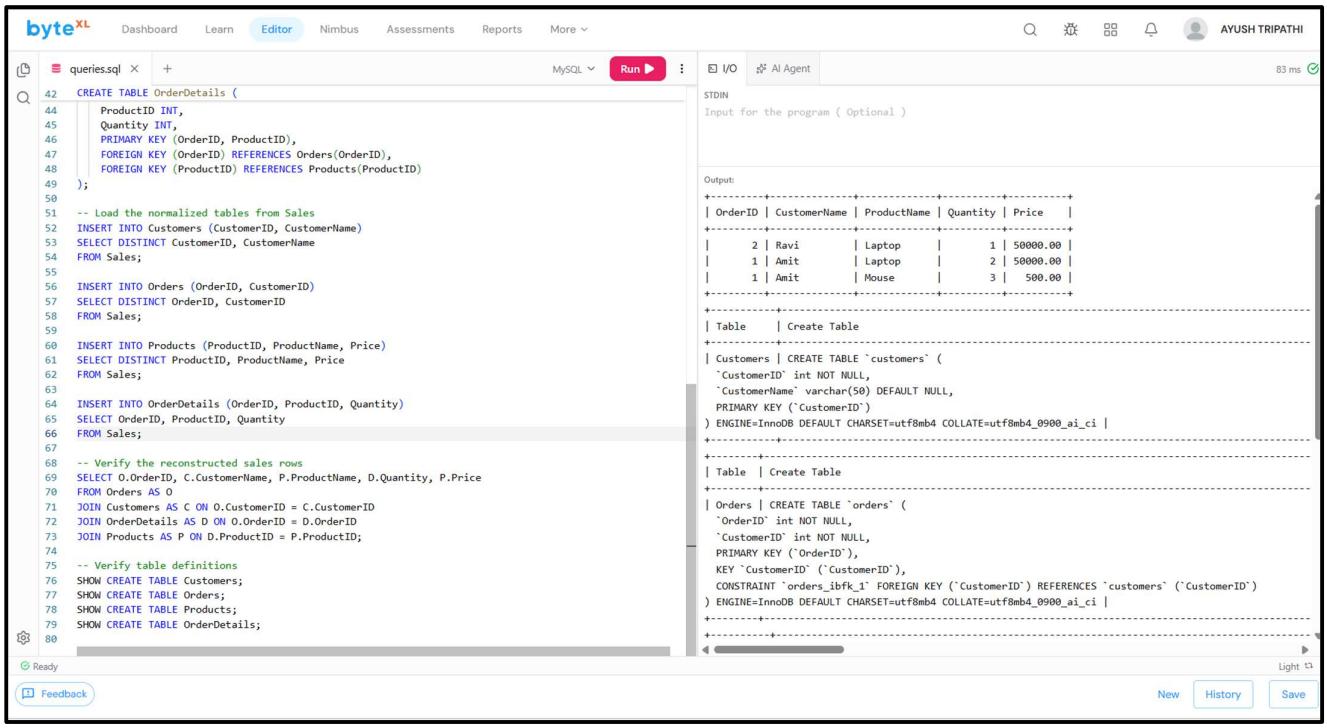
INSERT INTO Products (ProductID, ProductName, Price)
SELECT DISTINCT ProductID, ProductName, Price
FROM Sales;

INSERT INTO OrderDetails (OrderID, ProductID, Quantity)
SELECT OrderID, ProductID, Quantity
FROM Sales;

-- Verify the reconstructed sales rows
SELECT O.OrderID, C.CustomerName, P.ProductName, D.Quantity, P.Price
FROM Orders AS O
JOIN Customers AS C ON O.CustomerID = C.CustomerID
JOIN OrderDetails AS D ON O.OrderID = D.OrderID
JOIN Products AS P ON D.ProductID = P.ProductID;

-- Verify table definitions
SHOW CREATE TABLE Customers;
SHOW CREATE TABLE Orders;
SHOW CREATE TABLE Products;
SHOW CREATE TABLE OrderDetails;
```

OUTPUT :


byteXL
Dashboard
Learn
Editor
Nimbus
Assessments
Reports
More
AYUSH TRIPATHI
queries.sql
MySQL
Run
I/O
AI Agent
83 ms
STDIN
Input for the program (Optional)
Output:
CREATE TABLE OrderDetails (
ProductID INT,
Quantity INT,
PRIMARY KEY (OrderID, ProductID),
FOREIGN KEY (OrderID) REFERENCES Orders(OrderID),
FOREIGN KEY (ProductID) REFERENCES Products(ProductID)
);
-- Load the normalized tables from Sales
INSERT INTO Customers (CustomerID, CustomerName)
SELECT DISTINCT CustomerID, CustomerName
FROM Sales;
INSERT INTO Orders (OrderID, CustomerID)
SELECT DISTINCT OrderID, CustomerID
FROM Sales;
INSERT INTO Products (ProductID, ProductName, Price)
SELECT DISTINCT ProductID, ProductName, Price
FROM Sales;
INSERT INTO OrderDetails (OrderID, ProductID, Quantity)
SELECT OrderID, ProductID, Quantity
FROM Sales;
-- Verify the reconstructed sales rows
SELECT O.OrderID, C.CustomerName, P.ProductName, D.Quantity, P.Price
FROM Orders AS O
JOIN Customers AS C ON O.CustomerID = C.CustomerID
JOIN OrderDetails AS D ON O.OrderID = D.OrderID
JOIN Products AS P ON D.ProductID = P.ProductID;
-- Verify table definitions
SHOW CREATE TABLE Customers;
SHOW CREATE TABLE Orders;
SHOW CREATE TABLE Products;
SHOW CREATE TABLE OrderDetails;
OrderID	CustomerName	ProductName	Quantity	Price
2	Ravi	Laptop	1	50000.00
1	Amit	Laptop	2	50000.00
1	Amit	Mouse	3	500.00
Table	Create Table
Customers	CREATE TABLE `customers` (
`CustomerID` int NOT NULL,
`CustomerName` varchar(50) DEFAULT NULL,
PRIMARY KEY (`CustomerID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
Table	Create Table
Orders	CREATE TABLE `orders` (
`OrderID` int NOT NULL,
`CustomerID` int NOT NULL,
PRIMARY KEY (`OrderID`),
KEY `CustomerID` (`CustomerID`),
CONSTRAINT `orders_ibfk_1` FOREIGN KEY (`CustomerID`) REFERENCES `customers` (`CustomerID`)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
Ready
Light
Feedback
New
History
Save